

Beyond Steinhart-Hart: A Self-Calibrating Thermal Framework Using Dynamic Polynomial Extrapolation for Precise Battery Cell Monitoring

Leveraging Natural Thermal Cycles for Zero-Intervention Calibration

The foundational premise of this research is the development of a self-calibrating thermal architecture that requires no manual intervention by leveraging naturally occurring, slow thermal cycles. The core hypothesis—that a parked battery assembly experiences predictable and wide-ranging temperature sweeps—is strongly supported by real-world environmental data and established principles of thermal metrology. This approach, termed "Opportunistic Environmental Calibration," transforms passive data logging into an active, autonomous calibration process, turning the vehicle's own exposure to ambient conditions into a powerful tool for maintaining measurement fidelity. The feasibility of this method is rooted in the predictable diurnal temperature variations and extreme heat accumulation scenarios common in tropical climates like the Philippines. Manila, for instance, has an annual average mean temperature of 27.8°C, with monthly maxima reaching 31.6°C [571573](#). During peak summer months like May, the mean temperature can reach 29°C, further increasing the potential thermal excursion range [571](#). More critically, the scenario of a battery pack sitting in a closed metal box or car trunk exposed to direct sunlight provides an even more potent calibration opportunity. In such conditions, internal temperatures can easily exceed 50°C and potentially reach 55-60°C, creating a vast and well-defined thermal span ideal for multi-point calibration [174400](#).

The success of this opportunistic strategy hinges on the ability of the embedded firmware to accurately identify periods of "thermal equilibrium" from the continuous telemetry stream. A state of true equilibrium is defined as a condition where the battery pack is at zero or negligible load and all internal components have had sufficient time to thermally stabilize, resulting in minimal temperature gradients and a near-zero rate of temperature change across all sensors [123459](#). The firmware must implement robust logic to detect these windows. This involves monitoring several key metrics simultaneously: load current

must be below a predefined threshold (e.g., $< 200\text{mA}$ to account for parasitic standby loads), the derivative of the AHT20 temperature (dT_{AHT20}/dt) must be less than a small tolerance (e.g., $< 0.05\text{ }^{\circ}\text{C}/\text{min}$), and the spread between all eight NTC channels must be within a tight bound (e.g., $< 0.1^{\circ}\text{C}$) [455](#). By identifying these stable plateaus during both heating and cooling phases of the natural cycle, the firmware can harvest thousands of (NTC Resistance, AHT20 Temperature) data points. This dataset becomes the foundation for the calibration algorithm, effectively creating a custom calibration curve for each individual NTC channel that is specific to its unique electrical and thermal environment. This methodology mirrors established practices in laboratory calibration, which emphasize the importance of stabilizing the device under test before taking measurements to ensure accuracy [408590](#). The key innovation here is replacing the controlled laboratory bath with the uncontrolled but highly informative thermal environment of the parked vehicle.

This approach offers a significant advantage over traditional methods that rely on infrequent, manually-initiated calibration events or generic thermistor tables derived from datasheets. Datasheet values represent nominal characteristics and do not account for manufacturing tolerances, component aging, or the specific thermal coupling of the sensor to the cell [6 176](#). By performing a full thermal sweep—from a cool morning start (e.g., 24°C) through a hot afternoon (e.g., 52°C) and back down in the evening—the system captures the NTC's resistance-temperature behavior across its entire intended operating range [576](#). This comprehensive dataset allows for a much more accurate and physically representative calibration than would be possible with just three isolated points. Furthermore, the system's design inherently accounts for the thermal mass of the battery pack itself. Unlike a surface-mounted sensor on a small PCB, the NTC beads bolted to large cells and the co-located AHT20 share a complex thermal path involving the cell casing, thermal interface material, and the surrounding pack structure [291438](#). The natural thermal cycles provide ample time for these various masses to equilibrate, ensuring that the measured temperatures from the NTCs and the AHT20 are indeed representative of the same underlying thermal state of the assembly [612](#). This makes the AHT20 a valid reference point for calibrating the NTCs, despite not being in direct contact with the cell interiors. The system's reliance on real-world physics rather than idealized models is its greatest strength, enabling it to build a robust and reliable thermal model without any user action.

A Hybrid Calibration Framework: Maximizing Precision with Look-Up Tables and Dynamic Polynomials

To achieve the highest possible precision while ensuring robust extrapolation, the proposed system employs a sophisticated hybrid calibration framework. This architecture intelligently combines the strengths of a dense Look-Up Table (LUT) for precise interpolation within the empirically calibrated range and a continuously updated fourth-order polynomial for physically plausible extrapolation beyond those bounds. This dual-model approach directly addresses the limitations of relying on a single analytical equation, such as the conventional Steinhart-Hart (S-H) formula, which can exhibit significant errors when applied outside the range of its original calibration points [26](#). The LUT component is generated from the wealth of data collected during the opportunistic thermal cycles. For each NTC channel, the firmware stores (Resistance, Temperature) pairs corresponding to verified equilibrium states. These pairs form the basis of the LUT. Instead of simple linear interpolation, using a higher-order method like cubic spline interpolation ensures a smooth and highly accurate temperature estimate within the calibrated range [469](#). This method eliminates "model error"—the discrepancy between the true physical R/T curve and the mathematical approximation—which is a primary source of uncertainty in analytical equations [289](#). Given the high resolution of the INA226 ADC and careful circuit design, this LUT-based interpolation can readily achieve a precision of 0.01°C to 0.05°C, which is more than sufficient for detecting subtle cell-to-cell differences indicative of a developing fault [513539](#).

For temperatures outside the calibrated LUT range, the system relies on a dynamically updated fourth-order polynomial model. This model serves as the "physics engine" for extrapolation. The general form of this polynomial, often referred to as the Hoge equation, is expressed as $1/T = A_0 + A_1 \ln R + A_2 (\ln R)^2 + A_3 (\ln R)^3 + A_4 (\ln R)^4$ [532](#). Research has shown that such higher-order polynomials can outperform the standard three-parameter S-H equation, especially over wide temperature spans [528532](#). To prevent the polynomial from producing nonsensical results (e.g., exploding to infinity), it is anchored using "phantom anchors." These are theoretical resistance-temperature points calculated from the manufacturer's nominal NTC parameters (e.g., Beta value) at the edges of the expected operating range (e.g., -20°C and 100°C) [439](#). The weights assigned to these phantom anchors are kept low, allowing the empirical data from the LUT to dominate the fit, while still constraining the polynomial's behavior at the extremes [510](#). The coefficients ($A_0, A_1, \dots, A_4$) are not static; they are recalculated by a background task on the ESP32 whenever new, statistically significant calibration data becomes available. This refit trigger is twofold: first, when a new LUT bin is populated for the first time,

expanding the known temperature range; and second, when the sample count within an existing bin crosses a power-of-ten threshold (e.g., from 499 to 500 samples), indicating a significant increase in statistical confidence [176](#). This continuous refinement ensures the extrapolation model adapts to any long-term drift or changes in the NTC's characteristics, providing a much more reliable prediction at the edge of the operating envelope than a simple beta model based on initial factory data. The implementation of this matrix inversion and regression on an Arm Cortex-M processor is computationally feasible, typically taking only milliseconds, making it suitable for execution in a non-blocking FreeRTOS task [273274](#).

The runtime operation of this hybrid model is straightforward and efficient. When a temperature reading is required, the firmware first determines if the measured NTC resistance falls within the bounds of the calibrated LUT. If it does, high-precision interpolated values from the LUT are returned. If the resistance is outside this range, the system seamlessly switches to evaluating the dynamically updated fourth-order polynomial. This architecture elegantly balances the need for high-fidelity interpolation, where a LUT excels, with the challenge of robust extrapolation, which benefits from a physically-constrained polynomial model. It represents a pragmatic and powerful solution for a BMS operating in a variable real-world environment, ensuring maximum precision where it is most needed while gracefully handling conditions at the limits of its operational capability. The following table summarizes the two components of this hybrid model.

Feature	Look-Up Table (LUT) + Interpolation	Fourth-Order Polynomial
Primary Use Case	High-precision interpolation within the empirically calibrated temperature range.	Robust extrapolation outside the calibrated range.
Data Source	Empirically measured (Resistance, Temperature) pairs logged during thermal equilibrium.	Weighted least-squares regression of all historical calibration data plus "phantom anchors" from the NTC datasheet.
Mathematical Form	Cubic spline interpolation between discrete data points.	$1/T=A_0+A_1\ln R+A_2(\ln R)^2+A_3(\ln R)^3+A_4(\ln R)^4$.
Strengths	Eliminates model error; provides the highest possible precision for interpolated values.	Physically constrained, preventing runaway extrapolation; adapts to new data via dynamic refitting.
Weaknesses	Provides no information outside its calibrated bounds; requires sufficient historical data.	Prone to model error within the calibrated range compared to a LUT; computationally slightly more intensive.
Implementation Trigger	Populating new LUT bins or significantly increasing sample counts within existing bins.	Continuous background task that updates coefficients based on the latest calibration data.

Achievable Precision: Relative vs. Absolute Accuracy in Real-World Operation

In the context of battery management, distinguishing between absolute accuracy and relative precision is paramount. The primary goal of this system is not to report the exact, ground-truth temperature of the environment, but to maximize the relative precision (ΔT) between individual cells to reliably detect anomalies [29](#). This focus on relative performance is correct, as safety and health diagnostics depend on identifying deviations from the norm—such as one cell running hotter than its neighbors—not on knowing the pack's temperature to the nearest hundredth of a degree. After implementing the hybrid LUT and polynomial calibration against the co-located AHT20 sensor, the achievable relative precision between the eight NTC channels is exceptionally high. The dominant limiting factor shifts from the thermistor's intrinsic non-linearity to the electrical noise floor of the measurement system and minor inconsistencies between the channels. With the INA226's 16-bit resolution (1.25 mV/LSB) and proper signal averaging, the raw voltage measurement noise can be reduced to a level corresponding to millikelvin-scale temperature changes around room temperature [113539](#). While systematic errors from the divider resistors and sensor self-heating remain, the calibration process largely removes the random and offset errors inherent in uncalibrated NTCs. Therefore, achieving a relative precision of **0.01°C to 0.05°C** between any two sensors is a realistic engineering target. This level of precision enables the detection of extremely subtle temperature differences, such as a 0.05°C deviation, which could indicate a nascent issue like a weak connection or a slight imbalance in cell chemistry, far earlier than systems with lower resolution might.

In contrast, the system's absolute accuracy—the degree to which the reported temperature matches the true physical temperature—is fundamentally capped by its weakest link: the AHT20 sensor. The AHT20 is specified to have a typical temperature accuracy of $\pm 0.3^\circ\text{C}$ [39 105110](#). This means that even after a perfect calibration against the AHT20, the entire system's temperature readings will inherit this $\pm 0.3^\circ\text{C}$ uncertainty. For example, if the AHT20 reads 30.0°C , the actual temperature of the battery assembly could be anywhere from 29.7°C to 30.3°C . The calibration process cannot improve upon this inherent limitation of the reference sensor. However, the calibration does significantly enhance the system's *repeatability*. A well-calibrated system will consistently read 30.0°C every time the assembly returns to that exact thermal state, even if that 30.0°C reading is offset from the true world temperature by $+0.2^\circ\text{C}$. This repeatability is crucial for tracking temperature changes over time. The table below compares the achievable precision for relative and absolute metrics.

Metric	Realistic Expectation	Primary Limiting Factor(s)
Relative Precision (ΔT)	0.01°C to 0.05°C	Electrical noise floor, MUX settling, channel-to-channel consistency.
Absolute Accuracy	$\pm 0.3^\circ\text{C}$ (Limited by AHT20 sensor)	AHT20 sensor accuracy specification.
Long-Term Repeatability	$< 0.1^\circ\text{C}$	Sensor stability, thermal contact consistency, resistor drift.
Detection Threshold for ΔT	$\sim 0.05^\circ\text{C}$	Limited by relative precision and noise floor.

It is also important to consider the difference between the temperature measured by the NTC bead on the cell surface and the internal temperature of the cell's active materials. Even with perfect NTC calibration, there will always be a thermal gradient between the surface and the core, caused by the finite thermal conductivity of the cell casing and the thermal interface material [230309](#). The magnitude of this gradient depends on the cell's state of charge, age, and the rate of heat generation (i.e., the load). Under light load or at rest, this gradient may be minimal, perhaps just a fraction of a degree. However, under high-current operation, it can become significant, potentially masking the true internal temperature from the surface-mounted sensor [169](#). The purpose of the calibration is to ensure that the NTC provides the most accurate *surface temperature* measurement possible, which is the best indicator of the cell's thermal state that is practically accessible. Advanced BMS designs may incorporate thermal models to estimate the internal temperature from the surface measurement, but that is an extension beyond the scope of this specific calibration architecture [79 124](#). For the stated goal of maximizing relative precision between surface-mounted sensors, the described system is exceptionally well-suited.

Long-Term Stability and Drift Management of Glass-Bead NTC Thermistors

The longevity and reliability of a self-calibrating system depend critically on the stability of its core sensing components. The choice of NTC N58 glass-coated bead thermistors is a strategic decision that confers significant advantages in long-term stability, which is essential for a zero-intervention system operating over a year or more. Glass-encapsulated thermistors are renowned for their exceptional durability and resistance to environmental degradation [178586](#). The hermetic glass seal provides complete protection against moisture, chemicals, and mechanical stress, ensuring that the sensitive ceramic semiconductor element remains unaffected by harsh conditions [45 46](#). This encapsulation is a key reason for their excellent long-term stability, with industry sources reporting

minimal drift and suitability for applications up to 300°C [2](#) [42](#). Studies analyzing the long-term drift of glass-bead thermistors have confirmed their superior stability compared to other types, with some showing drift rates as low as 0.546 mK/year under controlled conditions [1](#) [127](#). General guidelines suggest that high-quality products can maintain an annual drift within 0.1°C [354](#). This inherent stability is a cornerstone of the system's viability.

Given this high baseline stability, the primary driver for the system's annual self-calibration is not major component aging but rather the cumulative effect of minor, reversible factors. The most significant of these is the thermal cycling experienced throughout the battery's life. Repeated heating and cooling can induce micro-stresses in the sensor's mounting and adhesive, potentially altering the thermal contact resistance over time [217524](#). Vibration and shock from vehicle operation can also contribute to loosening of connections or degradation of thermal interface materials [177](#). These changes, though small, can introduce a slowly varying bias into the temperature readings. The opportunistic calibration, performed annually during extended parking in varied ambient conditions, is perfectly suited to detect and correct for these mounting-related drifts. By comparing the current equilibrium residuals (the difference between the calibrated NTC and the AHT20) against historical data, the system can quantify any shift. If a sensor's reading has permanently drifted by, for example, +0.2°C, the next thermal sweep will capture this offset and allow the calibration model to adjust for it. This creates a powerful feedback loop that maintains accuracy without requiring manual recalibration.

Furthermore, the very act of self-heating introduces a small, controllable error. NTC thermistors dissipate power when a current passes through their voltage divider, causing them to heat up slightly [358554](#). The amount of self-heating depends on the dissipation constant of the thermistor and the excitation power, which is determined by the divider resistor value and the pack voltage [297](#). For a typical setup with a 10kΩ NTC and a 10kΩ fixed resistor powered by a ~13.4V pack, the power dissipation is relatively low, but not zero [330](#). This self-heating is generally considered a systematic error because it is relatively constant for a given set of conditions. During the calibration phase, the self-heating is present and gets averaged into the calibration data, effectively becoming part of the sensor's characteristic. As long as the excitation conditions (pack voltage, divider resistor) remain stable between calibration and operation, this self-heating effect largely cancels out when calculating ΔT between sensors. However, if the thermal environment changes dramatically between a calibration event and subsequent operation (e.g., from still air on a shelf to forced-air cooling during driving), the rate of heat dissipation from the sensor can change, leading to a variation in the self-heating-induced offset [222](#). To

mitigate this, the system design should aim for the lowest possible excitation current that still yields a good signal-to-noise ratio, minimizing the magnitude of the self-heating artifact [554](#). The combination of inherently stable glass-bead thermistors and an opportunistic calibration strategy that accounts for mounting effects and minor environmental changes makes this system remarkably resilient to long-term drift, fulfilling the requirement for a robust, zero-user-intervention solution.

Critical Edge Cases and Firmware-Level Mitigation Strategies

While the opportunistic calibration framework is robust, its effectiveness depends on the firmware's ability to navigate and mitigate several critical physical edge cases that can corrupt the calibration data. The most significant of these is the disparity in thermal response times between the NTC beads and the AHT20 sensor. NTC glass beads have extremely low thermal mass, giving them very fast response times, often in the millisecond range [44 476](#). In contrast, the AHT20 is typically an SMD component housed in a larger plastic or sintered-metal package, which acts as a thermal insulator and gives it a much slower response, with a specified thermal response time of 5 to 30 seconds [38 113](#). During a rapid temperature transition, such as moving the vehicle from a shaded area to direct sunlight, the NTC will register the temperature change almost instantaneously, while the AHT20 lags behind. If this thermal lag is ignored, the firmware will pair a slightly cooler NTC reading with a warmer AHT20 reading at a given timestamp, introducing a systematic error into the calibration curve. The primary mitigation strategy is to enforce a sufficiently slow thermal ramp rate for calibration eligibility. By setting a strict gate condition, such as $|dT/dt| < 0.05 \text{ }^{\circ}\text{C}/\text{min}$, the system ensures that it only collects data during periods of quasi-equilibrium where the transient lag effects are minimized [30](#). A more advanced firmware technique involves post-processing the data to actively align the signals. By calculating the time derivative of both the NTC and AHT20 temperature curves and finding the time shift that maximizes their cross-correlation, the firmware can determine the precise lag and shift one of the datasets to synchronize the thermal wavefronts before fitting the calibration model [347](#). This synchronization is crucial for achieving high-fidelity calibration during any unavoidable rapid transitions.

Another critical edge case is asymmetric self-heating. The power dissipated in an NTC, $P=I^2R$, generates heat that must be conducted away through the sensor's mounting [358](#). If the thermal coupling is perfect and uniform, this self-heating will raise the sensor's

temperature above the cell surface temperature by a consistent amount. However, in practice, this coupling can be imperfect. Uneven application of thermal grease, air gaps, or adhesive degradation can create localized "hot spots" or trap heat, causing a particular NTC to read significantly higher than its neighbors even when they are at the same true temperature ⁶⁷. This effect is particularly problematic because it can mimic a genuine cell fault. Mitigation requires a multi-pronged approach. First, the firmware should operate the NTC dividers at the lowest possible excitation current that still produces a clean signal, thereby minimizing the self-heating power ⁵⁵⁴. Second, since this is a systematic error, it affects all sensors similarly under identical conditions. Therefore, when the system is in thermal equilibrium (e.g., during a parking event), the calibration process will measure and lock in this self-heating offset. The primary consequence is a small, constant addition to the absolute temperature reading, which is often acceptable given the system's focus on relative precision (ΔT). Third, the system's validation capabilities can be used to detect sudden changes. A gradual increase in self-heating over years due to mounting degradation might go unnoticed, but a sudden change in the residual error after a load event could indicate a mounting problem that needs attention ⁴⁵⁵.

Finally, the stability of the fixed resistors in the voltage dividers is a critical consideration. The accuracy of the calculated NTC resistance, $R_{NTC} = R_{fixed} \frac{V_{NTC}}{V_{PACK} - V_{NTC}}$, is directly dependent on the precision and stability of R_{fixed} ³³⁰. Standard carbon film resistors can have tolerances of 1% or worse and temperature coefficients as high as 250 ppm/°C, meaning a 10°C temperature difference across the PCB could cause a 0.25% resistance change, translating to a noticeable temperature error ⁴⁴. To mitigate this, the design must specify the use of high-precision, low-temperature-coefficient resistors (e.g., 0.1% tolerance, 25 ppm/°C or better). Furthermore, all eight fixed resistors should be placed in close proximity on the PCB, ideally in a thermally stable location away from major heat sources like MOSFETs or voltage regulators. This minimizes thermal gradients across the resistors, ensuring they experience the same temperature change and thus canceling out the common-mode error in the differential voltage measurement. By addressing these three edge cases—sensor lag, asymmetric self-heating, and resistor instability—the firmware can transform raw, noisy sensor data into a highly reliable and precise thermal model.

System Behavior Over Time: From Annual Recalibration to Self-Healing Diagnostics

The long-term behavior of this self-calibrating system evolves from a simple annual recalibration process into a sophisticated, autonomous diagnostic and health-monitoring tool. The system's intelligence lies not just in its ability to perform a one-time calibration, but in its capacity to learn, adapt, and validate its own integrity over a full year of operation. The annual recalibration loop begins with the accumulation of calibration data during routine vehicle parking. Each day the vehicle sits idle, especially during periods of significant ambient temperature change, the firmware opportunistically adds valuable data points to the calibration database. A week-long period of high ambient temperatures, for example, provides a dense cloud of data points at the upper end of the operating range (e.g., 40-50°C), which is precisely where the extrapolation model is most critical. The background firmware task continuously refits the fourth-order polynomial, progressively tightening its constraints as more data is incorporated. This dynamic updating ensures that the extrapolation model remains physically grounded and accurate, adapting to any minute long-term drift in the NTCs' characteristics [176](#). This continuous improvement loop is the essence of the system's self-calibration capability.

Beyond recalibration, the system's most powerful feature is its self-healing and anomaly detection capability, enabled by the AHT20 sensor. After a period of dynamic load, when currents are high and heat generation is significant, the NTCs will read substantially higher temperatures than the AHT20, which measures the cooler assembly and ambient air temperature [86](#). This difference, $\Delta T_{NTC-AHT20}$, is not a sign of failure but a physically meaningful observation of the thermal separation between the hot cells and the reference sensor [203](#). The system learns this relationship under various load conditions to build a "sanity envelope" [135](#). However, the true diagnostic power is unlocked when the load ceases and the pack returns to thermal equilibrium. At this point, the NTCs and the AHT20 should converge again, and the residual error between them should return to its normal, stable value. The firmware can monitor this convergence closely. If, after multiple load events, the system observes that the equilibrium residual for a particular NTC has shifted permanently—for instance, it now settles at +0.1°C instead of +0.05°C—this indicates a potential change in the system's thermal characteristics [455](#). Such a shift could be caused by a number of issues, including a failing sensor, a change in thermal contact due to adhesive degradation or vibration, or even a manufacturing defect that only manifests under certain conditions [217218](#). By flagging these persistent shifts in the calibration residuals, the system moves beyond mere temperature measurement into

active health monitoring. It can generate alerts for maintenance, prompting a check of the physical mounting of the sensor or even initiating a forced recalibration sequence.

This autonomous diagnostic loop transforms the BMS from a passive safety device into an active observer of its own health. The system's ability to correlate changes in its calibration model with physical events (like load cycles and ambient temperature changes) provides a rich dataset for understanding the long-term behavior of the battery pack. It can detect the onset of problems long before they become critical, such as the early signs of a micro-short in a cell or a failing tab weld, which would manifest as a cell that heats up disproportionately during load ²⁹. The integration of a high-stability glass-bead NTC with a reliable AHT20, combined with intelligent firmware capable of opportunistic calibration and self-validation, creates a truly resilient and intelligent thermal management system. It is a system that not only meets the immediate need for high-precision relative temperature measurement but also builds a foundation for predictive maintenance and enhanced long-term safety, fully realizing the vision of a zero-user-intervention thermal observatory.

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